

Smoking–NRF2 Immune Dichotomy: Intrinsic vs Exogenous NRF2 Activation

Phase 1: NRF2 is independently prognostic (OS 4.14 vs >7.5 years) in NON–SMOKERS, but not in smokers (P = 0.569).
Hypothesis: Intrinsic NRF2 activation (non–smokers, likely NFE2L2/KEAP1 mutations) produces deeper immune exclusion than tobacco–driven NRF2 induction (smokers).
Test: Is Cliff's d for immune exclusion (NRF2–High vs Low) larger in non–smokers?

